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Artificial Intelligence

CSE-316

(Section: 65-D/H)

Instructor

Dr. M. Akhtaruzzaman

Associate Professor, CSE, DIU

Associate Professor and Head, RME, DIU

akhtaruzzaman.cse@diu.edu.bd;

akhter900@gmail.com

Theory Lecture-13

ANN Basic

02. 04, 2026, DIU, Dhaka.

رب زدني علما وارزقني فهما.

My Lord! Advance me in Knowledge and True Understanding.

Content

- ▶ Artificial Neural Network (Basic)
- ▶ Neuro Fuzzy System

Artificial Neural Networks

Artificial Neural Networks

What is ANN?

A collection of networked artificial neurons inspired by the structure of the brain – cerebral cortex – **used as a data processing system.**

What is artificial neuron?

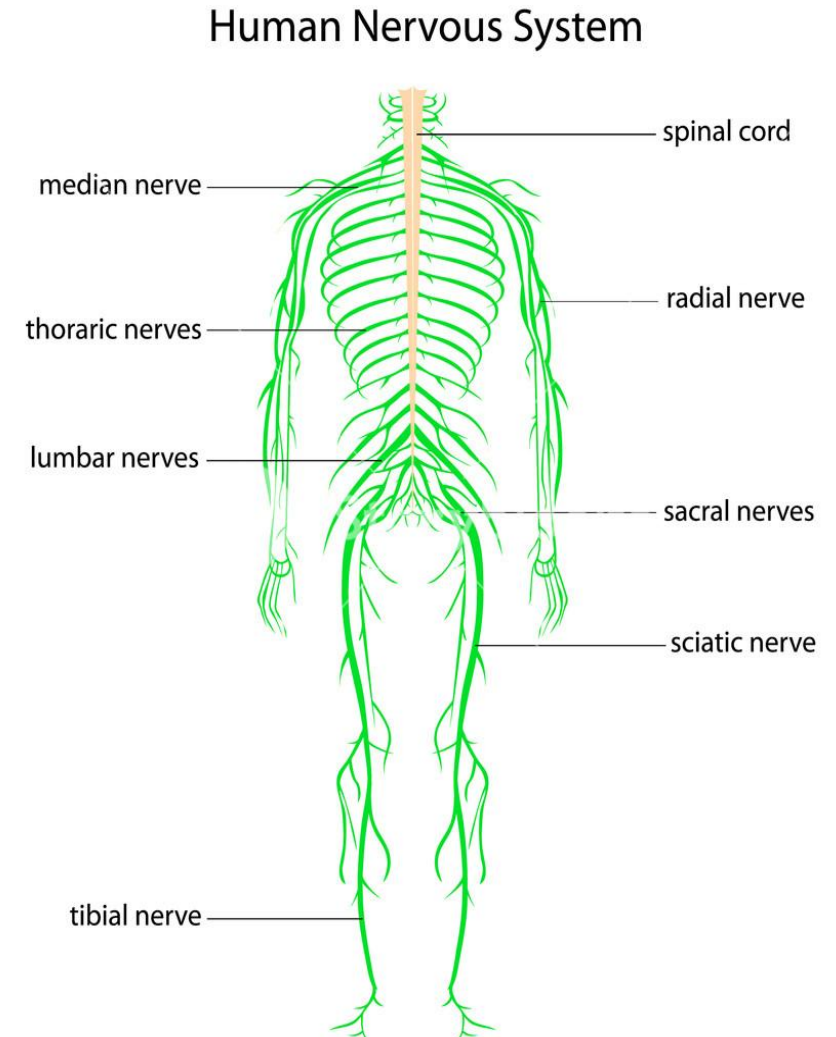
Artificial neuron is a mathematical model of biological neuron.

Human Nervous System

Human nervous system is the part of the human body that **coordinates a person's voluntary (conscious responses), involuntary, and Reflex actions** and **transmits signals between different parts of the body.**

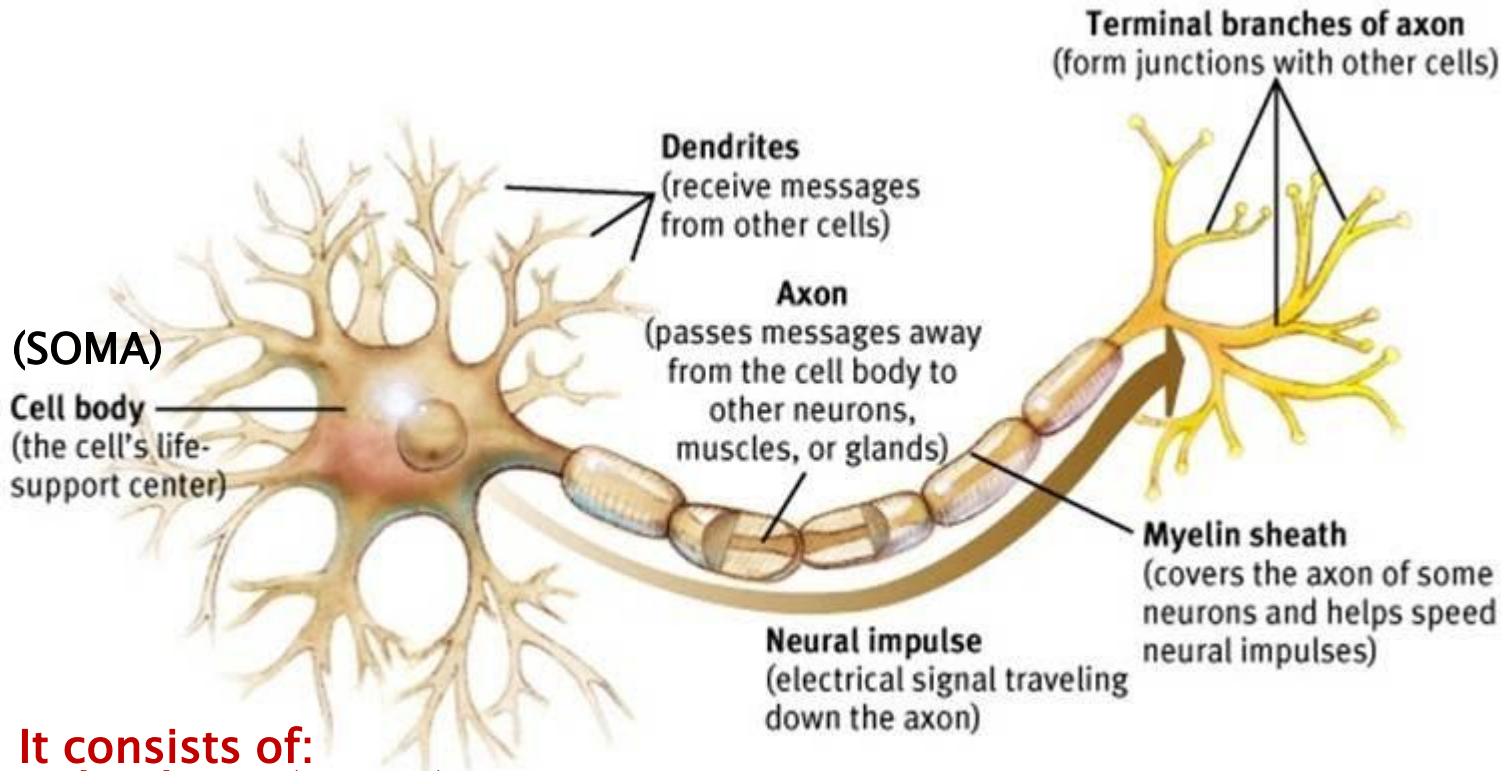
The human nervous system consists of two main parts,

- central nervous system (CNS)
- peripheral nervous system (PNS).



Biological Neuron

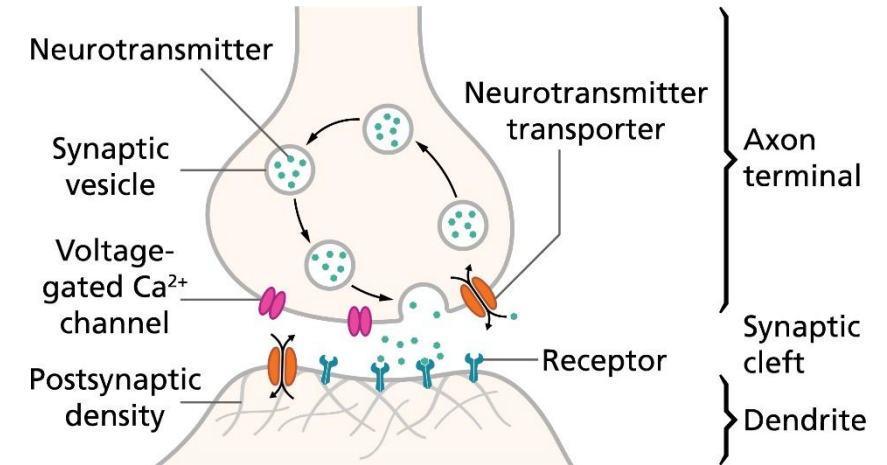
Biological neuron – basic element in nervous system



- It consists of:
- dendrites (inputs)
 - cell body (Soma)
 - axon (output)
 - synapse

A neuron collects signals and sums all the 'excitatory and inhibitory' influences acting on it.

The **excitatory** dominant the neuron 'fires' and sends that signal to another neuron through the synapses.



Dendrites receive signal from other neuron through **synapses** and pass them on to axon to be transmitted to another neuron.

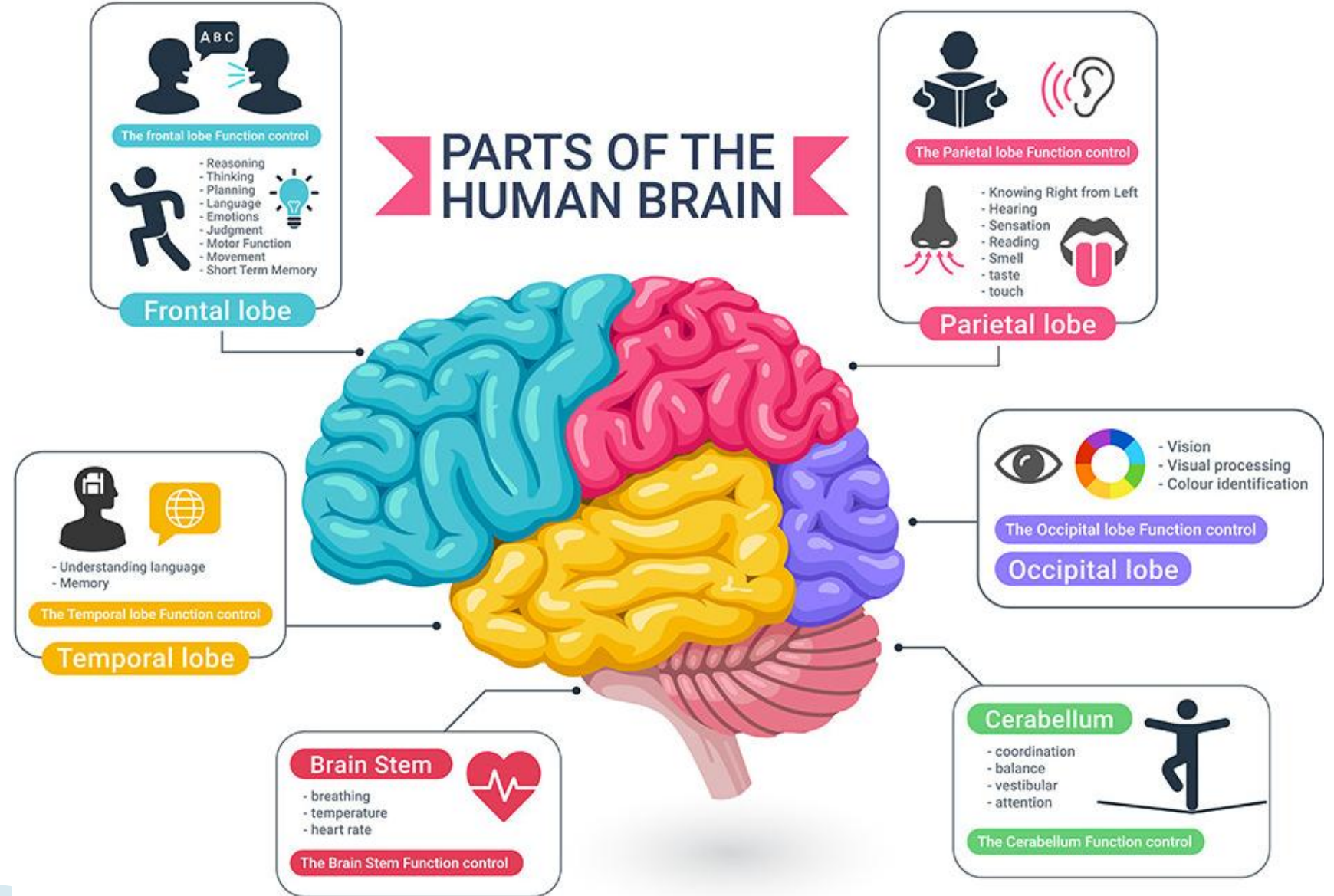
Synapse is the region at which a neuron communicates with another.

Biological Neural Network

Human brain

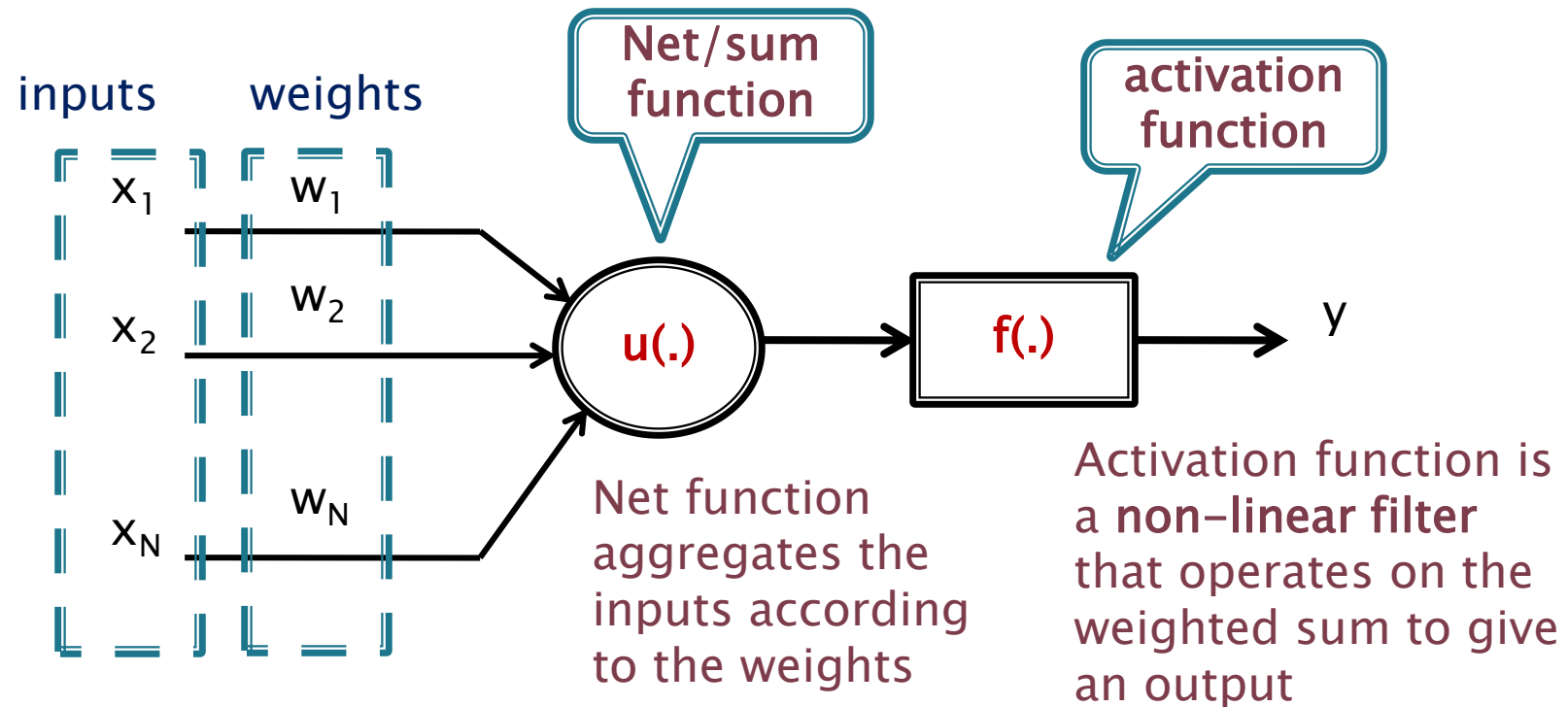
The human brain contains many neurons and synapses, estimated to be

10^{11} neurons
&
 10^{14} synapses



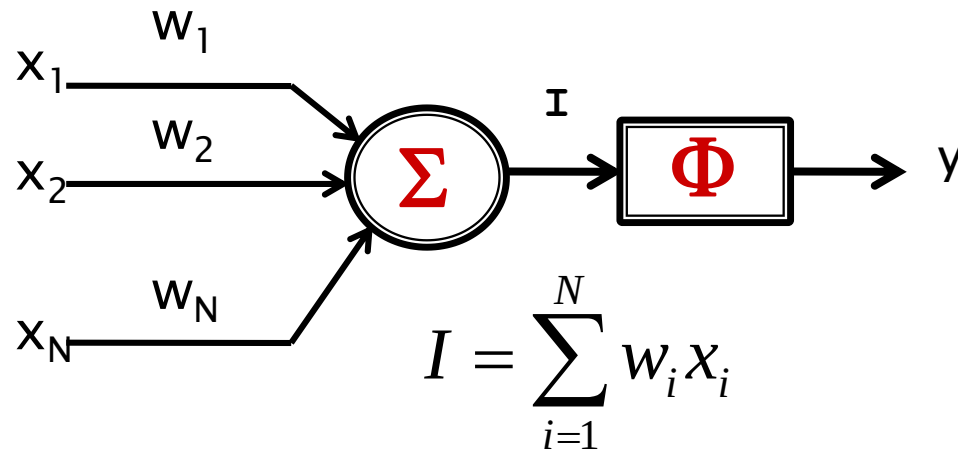
Artificial Neuron

Artificial neuron is a model whose components are analogs of an actual neuron



Artificial Neuron...

Net function and activation function



$$y = \phi(I - T)$$

T is the threshold value set by the activation function. Thus,

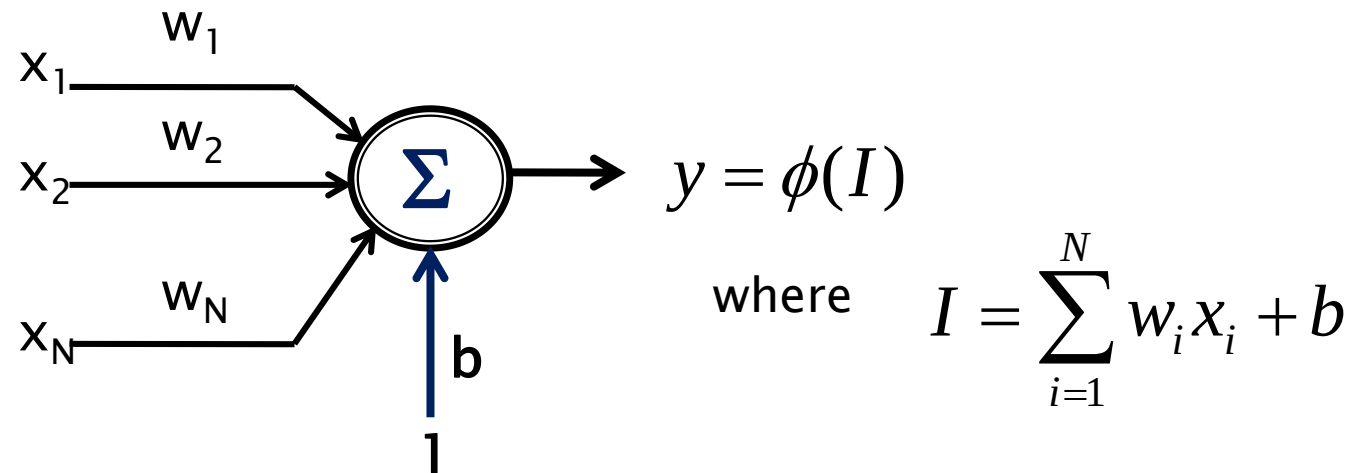
$$y = \begin{cases} -1 & I < T \\ +1 & I \geq T \end{cases}$$

Artificial Neuron...

Net function and activation function...

Bias or artificial threshold

The **threshold** can also be **replaced** by a constant **input value = 1** with a weight of **b** that is equal to the threshold.

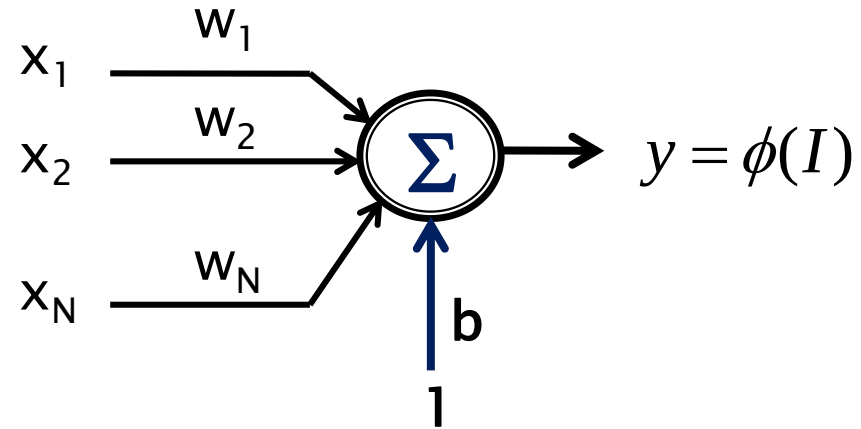


Artificial Neuron...

Net function and activation function...

Note that we can write the inputs and weights in vector/matrix notation, thus

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \quad \mathbf{W} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{bmatrix}$$



then

$$I = \sum_{i=1}^N w_i x_i = \mathbf{W}^T \mathbf{X}$$

and

$$y = \phi(\mathbf{W}^T \mathbf{X})$$

Note that for a single neuron, the output is a scalar quantity

Artificial Neuron...

Net function and activation function...

Another forms of net function is the quadratic function, but it is seldom used

$$I = \sum_{i=1}^N w_i x_i^2$$

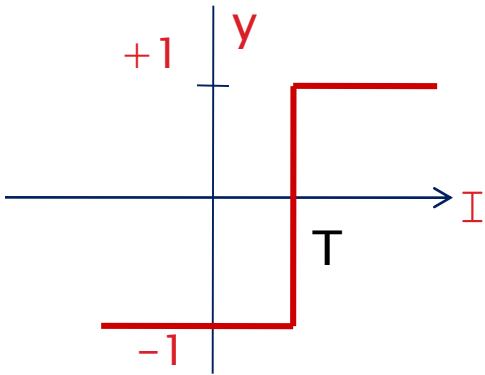
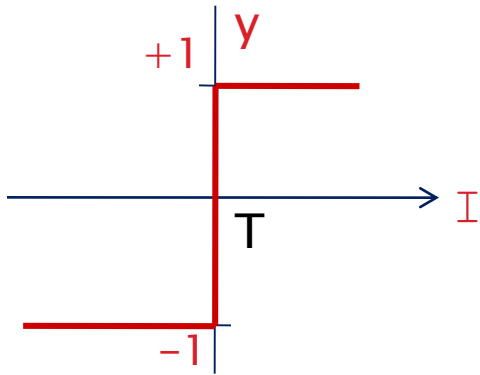
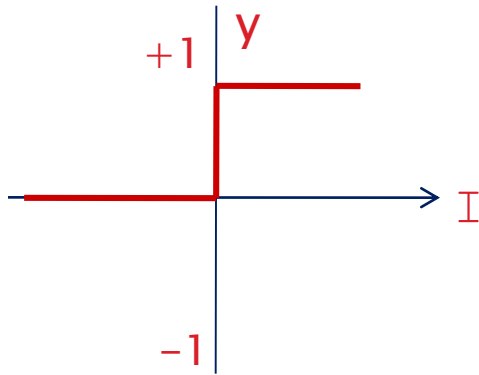
Types of activation function

1. Threshold
2. Signum
3. Step
4. Ramp
5. Unipolar sigmoid
6. Bipolar sigmoid

Artificial Neuron...

Net function and activation function...

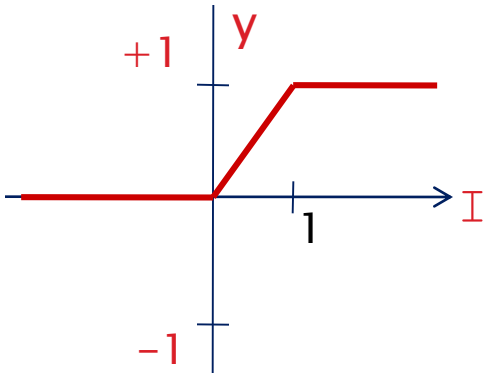
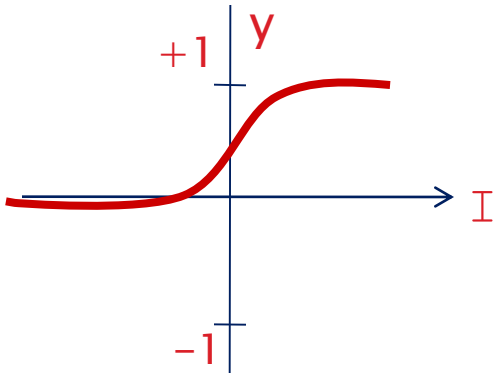
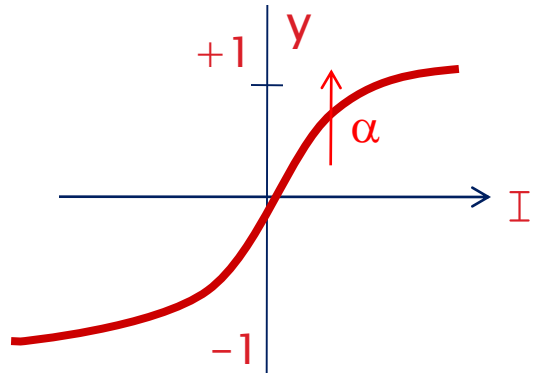
Types of activation function...

Threshold	Signum	Step
$y = \Phi(I) = \begin{cases} +1 & I \geq T \\ -1 & I < T \end{cases}$	$y = \Phi(I) = \begin{cases} +1 & I \geq 0 \\ -1 & I < 0 \end{cases}$	$y = \Phi(I) = \begin{cases} +1 & I \geq 0 \\ 0 & \text{otherwise} \end{cases}$
		

Artificial Neuron...

Net function and activation function...

Types of activation function...

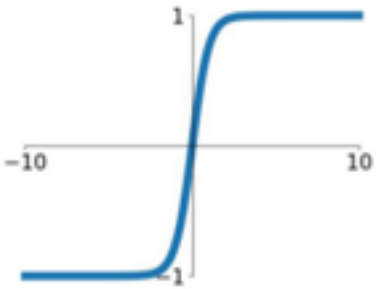
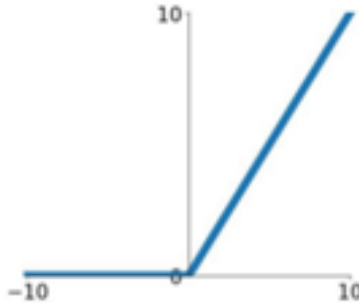
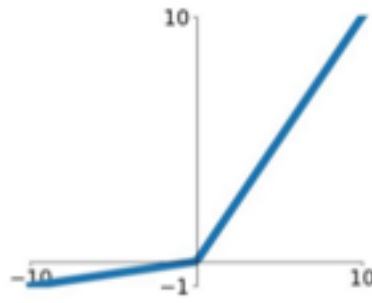
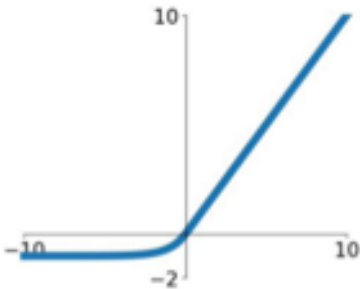
Ramp	Unipolar Sigmoid	Bipolar Sigmoid
$y = \Phi(I) = \begin{cases} 1 & I > 1 \\ I & 0 < I \leq 1 \\ 0 & I < 0 \end{cases}$	$y = \Phi(I) = \frac{1}{1 + e^{-\alpha I}}$	$y = \Phi(I) = \frac{2}{1 + e^{-\alpha I}} - 1$
α is slope of the sigmoid		
		

Artificial Neuron...

Net function and activation function...

Some more types of activation function...

Exponential Linear Unit

tanh	ReLU	Leaky ReLU	ELU
$\tanh(x)$	$\max(0, x)$	$\max(0.1x, x)$	$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$
			

Rectified Linear Unit

Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

Artificial Neuron...

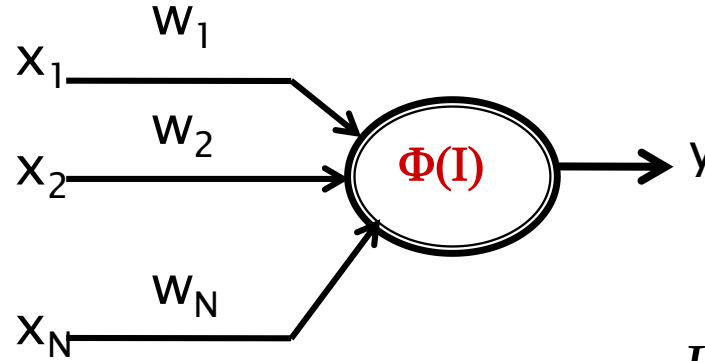
Example:

$$\Phi(I) = \begin{cases} 1, & \text{if } I \geq 0 \\ -1, & \text{if } I < 0 \end{cases}$$

Find y when

$w = [1 \ -1 \ 0.5]^T$, and

$x = [1 \ -2 \ 0.5]^T$



$$I = w^T x = \begin{bmatrix} 1 & -1 & .5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ .5 \end{bmatrix} = 3.25$$

$$y = \Phi(I) = \Phi(3.25) = 1$$

Artificial Neural Network (ANN)

ANN is a **data processing system** comprising a **collection of artificial neurons networked in a structure** that is inspired by and modeled after the human brain.

ANN is characterized by

1. The model of the neuron
2. The model of the synapses interconnection
3. Learning/Training rule

The model of the neuron is characterized by the **net/sum function** and the **activation function**.

Artificial Neural Network (ANN)...

ANN Connections

Is also called the structure or architecture of the ANN

It is characterized by the manner in which the neurons are interconnected

Two basic connections

1. Feed-forward network
2. Recurrent network

Artificial Neural Network (ANN)...

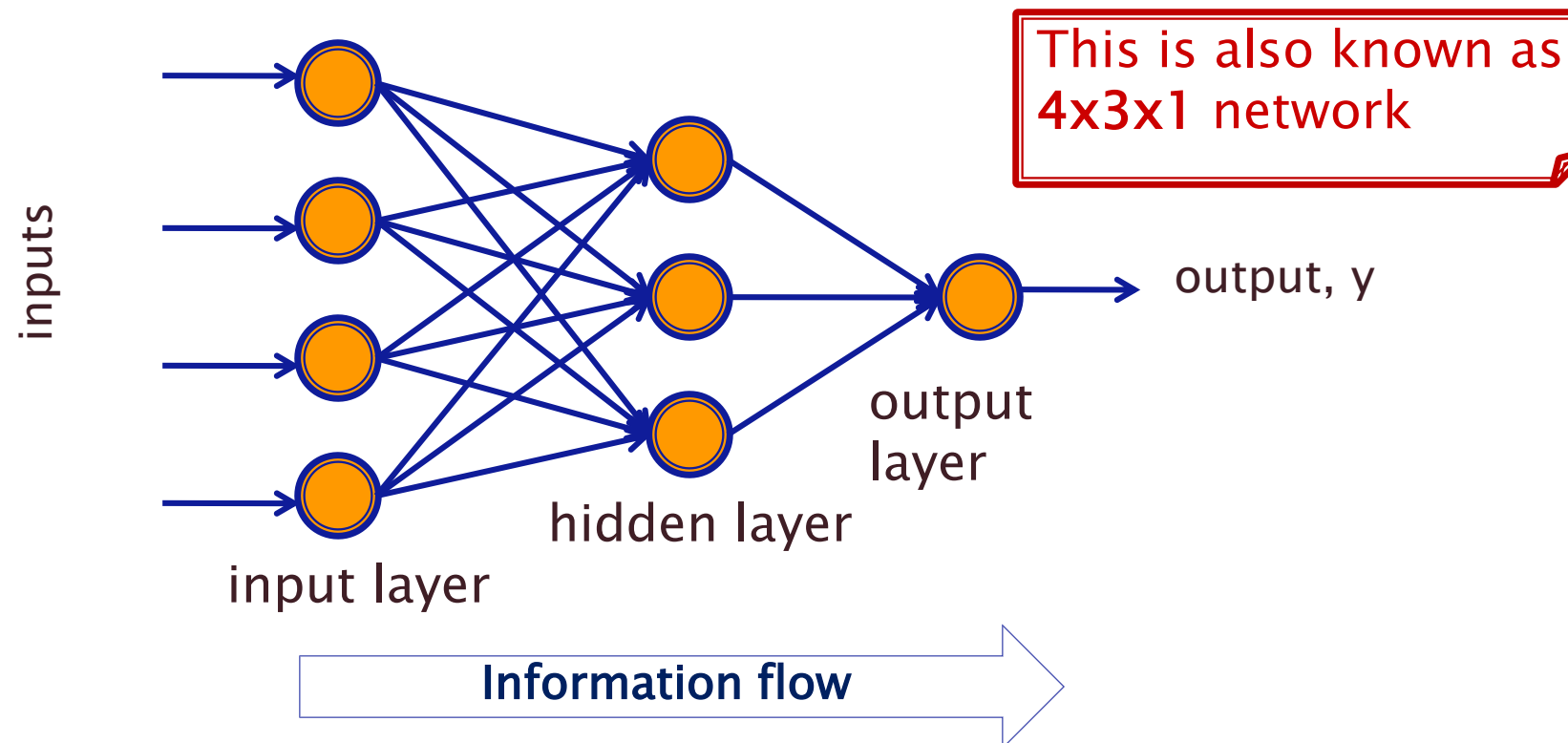
6 Types of Artificial Neural Networks Currently Being Used in Machine Learning:

- 1) Feedforward Neural Network
- 2) Radial basis function Neural Network
- 3) Kohonen Self Organizing Neural Network
- 4) Recurrent Neural Network (RNN) (Long Short Term Memory)
- 5) Convolutional Neural Network
- 6) Modular Neural Network

Artificial Neural Network (ANN)...

Feedforward Network

Neurons are arranged in one or more layers and information flows only in one direction; from the input to the output

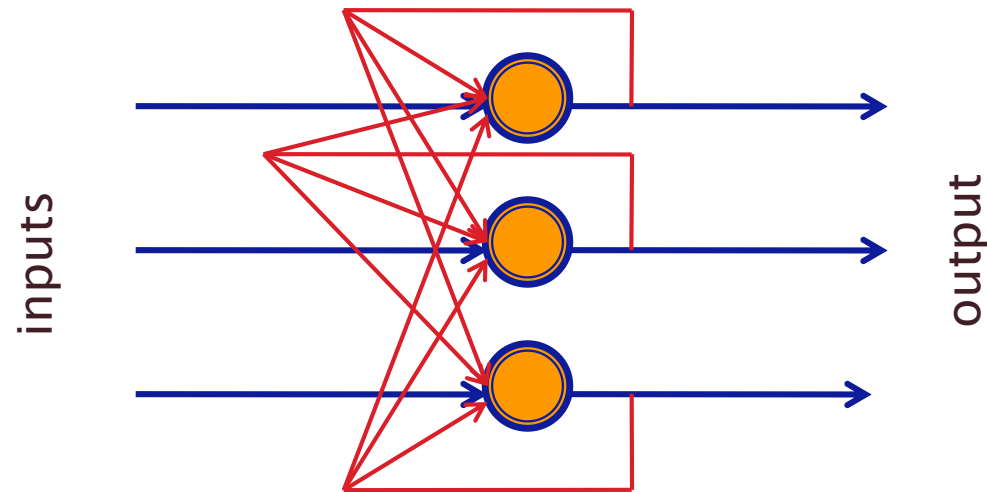


Artificial Neural Network (ANN)...

Recurrent Network

Neurons are arranged in one or more layers.

Feedback is introduced either **internally to the neuron** or **to another neuron in preceding layer**.



Single layer recurrent network

Artificial Neural Network (ANN)...

Learning/ Training Rule

ANN performs two main functions:

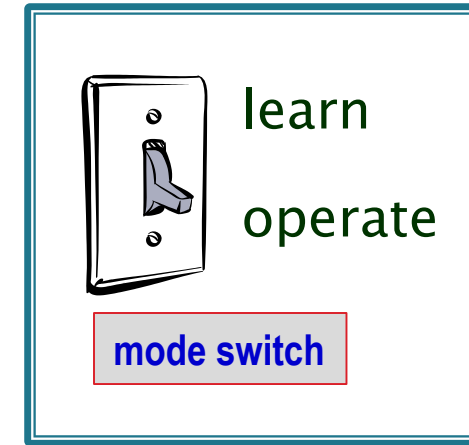
1. Learning

Learning is accomplished by **changing the network weights** to give desired output in response to stimulus input.

In biological neurons it is by changing the synaptic strengths, eliminating some synapses, and building new ones.

2. Recall

Is a process of accepting input stimulus and producing output response **in accordance with the network weights**



Artificial Neural Network (ANN)...

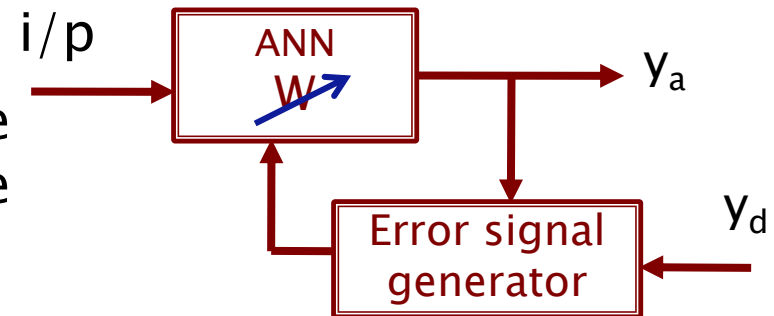
Learning/ Training Rule...

The aim of training is to ‘impart’ knowledge to the network so that it can give ‘correct’ solution when given ‘problems’ of similar nature in the area it was trained.

There are two main training approaches:

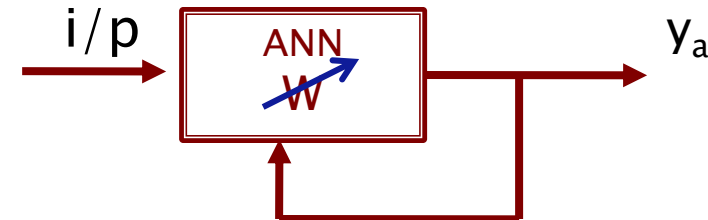
1. Supervised (with teacher)

Correct sets of output and input are ‘presented’ to the network and the weights are adjusted accordingly.



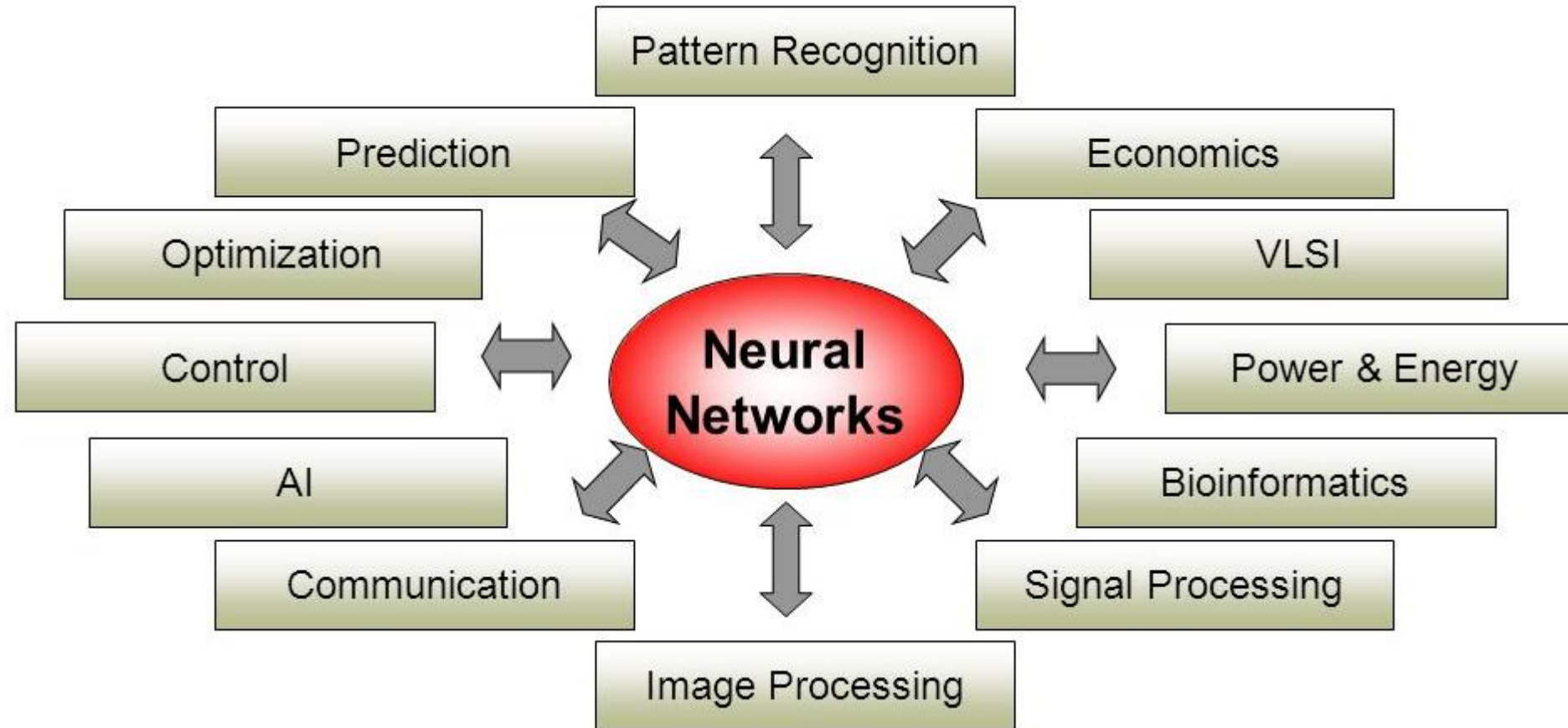
2. Unsupervised (without teacher)

The network learns by itself through ‘self-training’ algorithm.



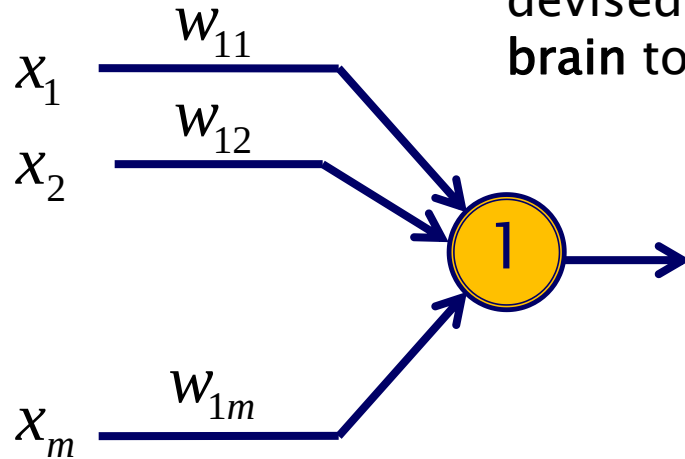
Artificial Neural Network (ANN)...

Applications



Perceptron

A computer model or computerized machine devised to **represent or simulate the ability of the brain** to recognize and discriminate.



1 threshold

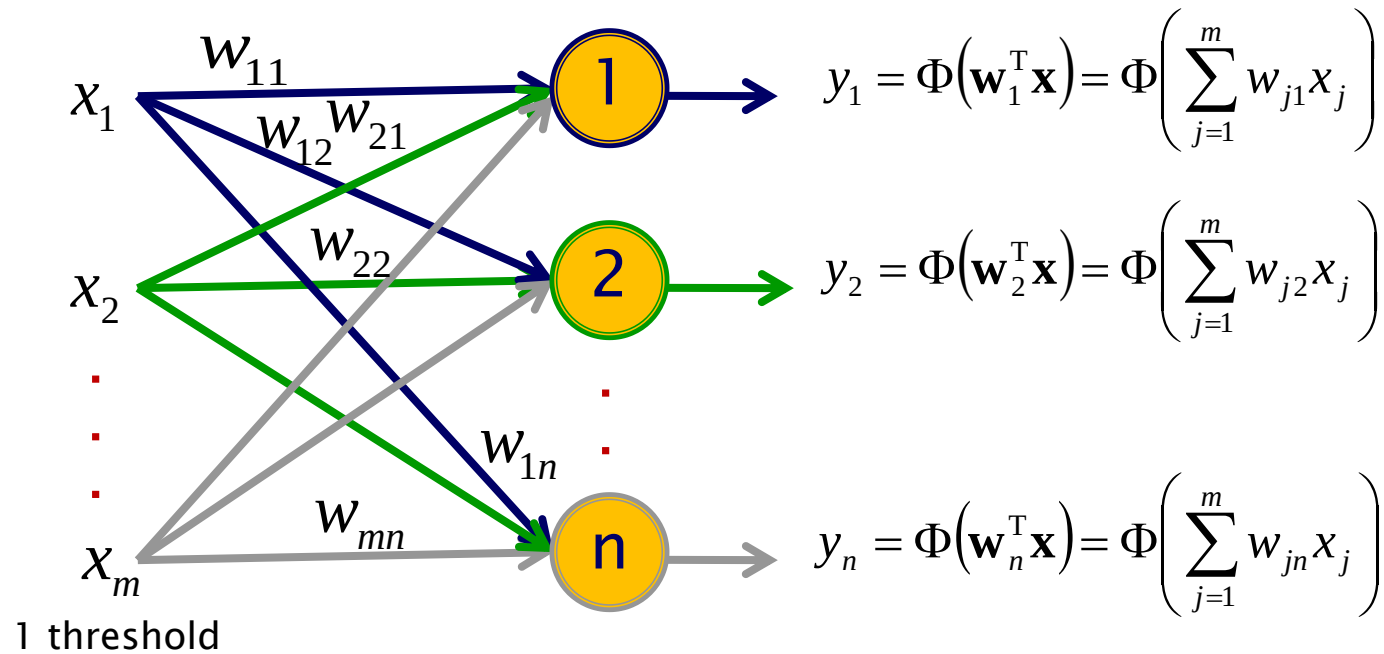
$$y_1 = \Phi(\mathbf{w}_1^T \mathbf{x})$$
$$= \Phi\left(\sum_{j=1}^m w_{1j} x_j\right)$$

m = number of inputs
n = number of neurons/nodes
For the first neuron/node

$$\mathbf{w} = \begin{bmatrix} w_{11} \\ w_{21} \\ \vdots \\ w_{m1} \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

Perceptron...

Single layer perceptron with m inputs, n nodes/neurons



In general,

$$y_i = \Phi(\mathbf{w}_i^T \mathbf{x}) = \Phi\left(\sum_{j=1}^m w_{ji} x_j\right)$$

Weight indices

$$w_{ij}$$

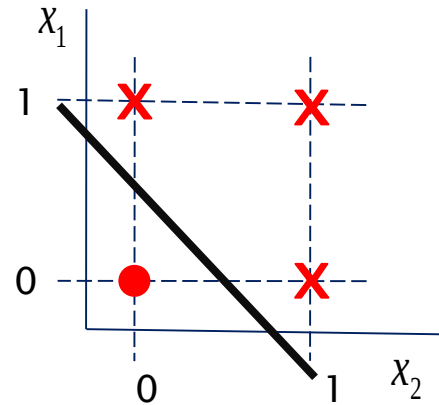
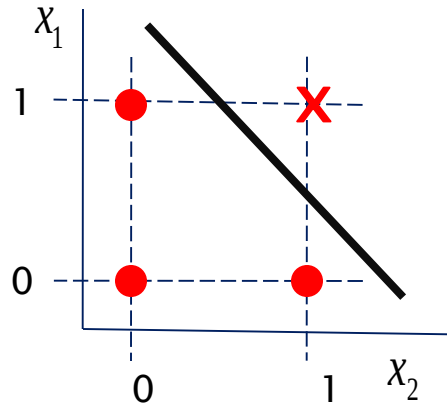
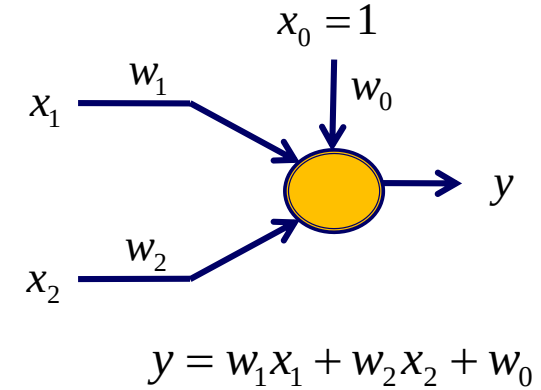
i= input index;
j= node index

Perceptron...

What can a perceptron do?

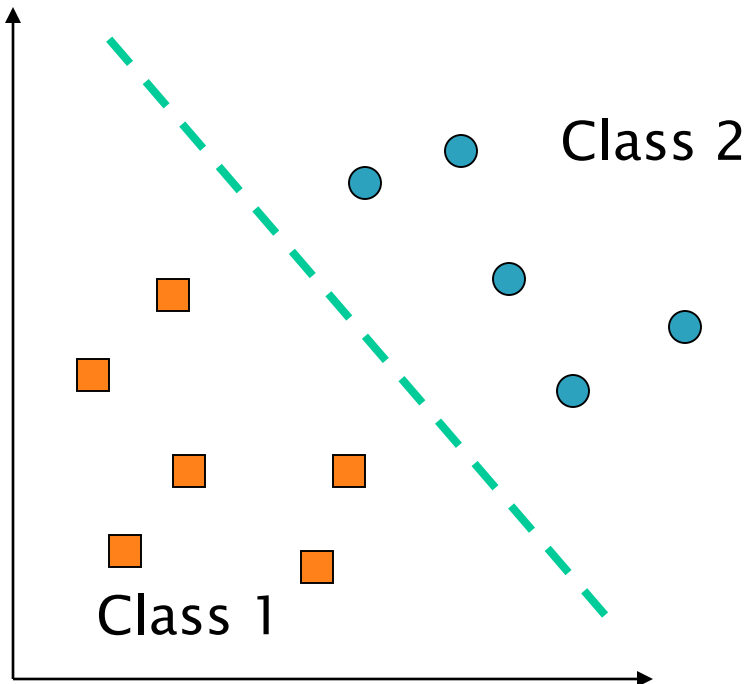
A single perceptron can **solve linear classification** problems.

Example: AND OR types classification by using step activation function

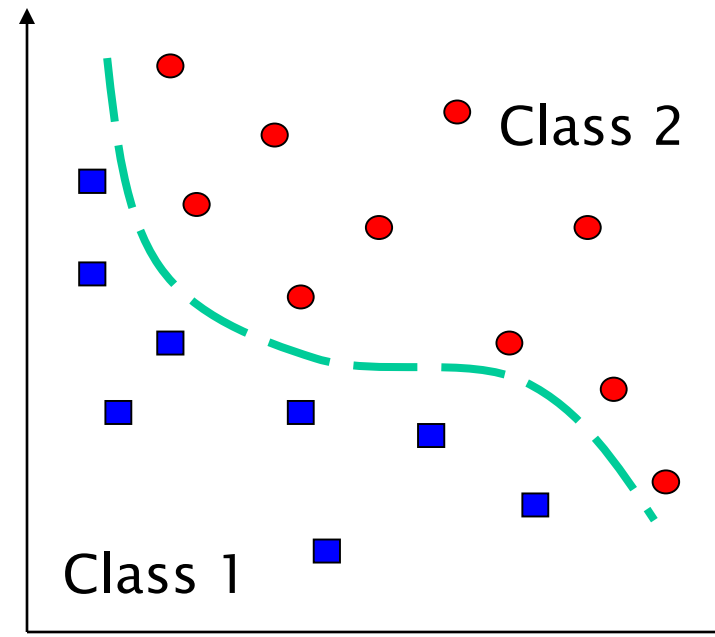


Perceptron...

Linearly Separable and Non-linearly Separable Problem



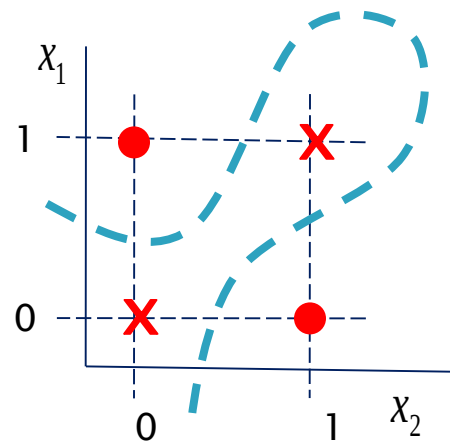
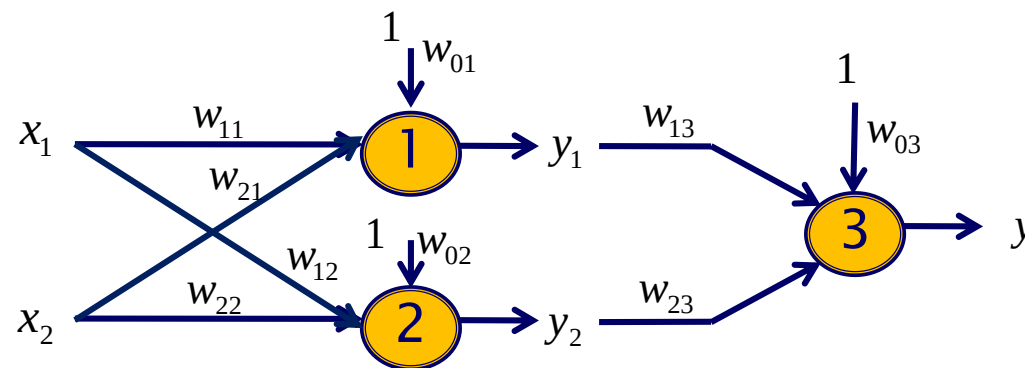
Linear Separable



Non-linear Separable

Perceptron...

Let's try two nodes for the EXCLUSIVE-OR classification



Perceptron...

$$y = w_{13}y_1 + w_{23}y_2 + w_{03}$$

$$y_1 = w_{11}x_1 + w_{21}x_2 + w_{01}$$

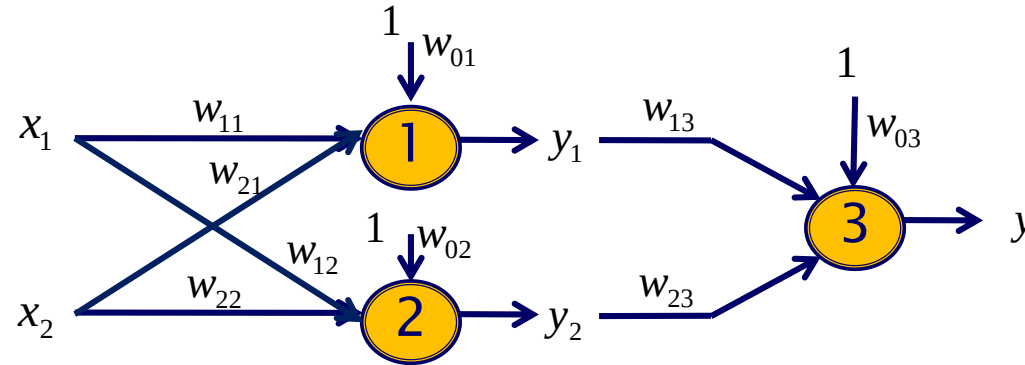
$$y_2 = w_{12}x_1 + w_{22}x_2 + w_{02}$$

Answer:

$$w_{11} = w_{13} = w_{23} = w_{22} = +1$$

$$w_{21} = -1; w_{12} = -1$$

$$w_{01} = -1; w_{02} = -1; w_{03} = -1$$



Adjust the weights so that the following outputs are satisfied for the layers

x_1	x_2	y_1	y_2
0	0	0	0
0	1	0	×
1	0	×	0
1	1	×	×

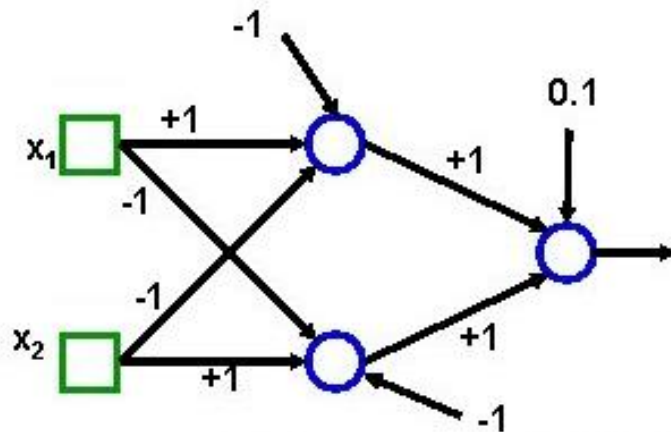
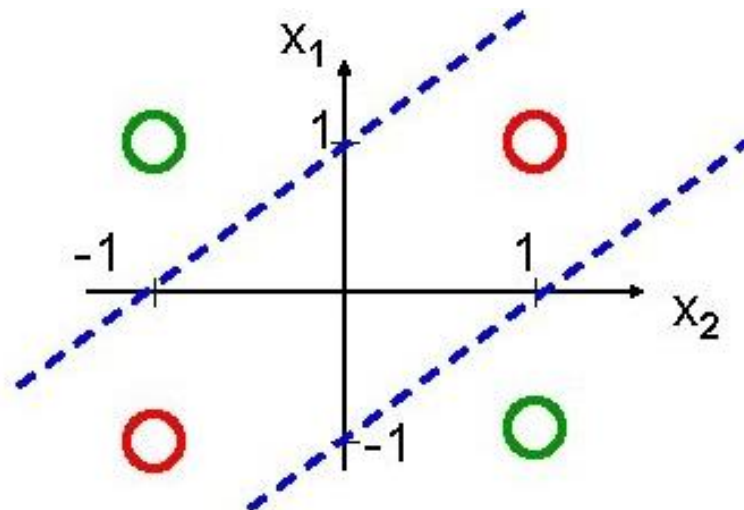
y_1	y_2	y
0	0	0
0	×	1
×	0	1
×	×	0

Convince yourselves that those are the answers

Perceptron...

An example of EXCLUSIVE-OR classification

x_1	x_2	$x_1 \text{ XOR } x_2$
-1	-1	-1
-1	1	1
1	-1	1
1	1	-1



$$\varphi(v) = \begin{cases} 1 & \text{if } v > 0 \\ -1 & \text{if } v \leq 0 \end{cases}$$

φ is the sign function.

Perceptron...

Code: EXCLUSIVE-OR Analytical Approach

```
clc;
clear all;

x = [0, 0;
     0, 1;
     1, 0;
     1, 1]

w11 = 1;
w12 = -1;
w21 = -1;
w22 = 1;
w13 = 1;
w23 = 1;
w01 = -1;
w02 = -1;
w03 = -1;

for i=1:1:4
    y1 = x(i,1)*w11 + x(i,2)*w21 + w01;
    % y1 activation...
    if(y1<0)
        y1 = 0;
    else
        y1 = 1;
    end
    y2 = x(i,1)*w12 + x(i,2)*w22 + w02;
    % y2 activation...
    if(y2<0)
        y2 = 0;
    else
        y2 = 1;
    end
    y3 = y1*w13 + y2*w23 + w03;
    % y3 activation...
    if(y3<0)
        y3 = 0;
    else
        y3 = 1;
    end
end
```

Output:

x =

0	0
0	1
1	0
1	1

y3 =

0

y3 =

1

y3 =

1

y3 =

0

Perceptron Learning Rule

Learning (setting or adjustment of weights) can be accomplished using

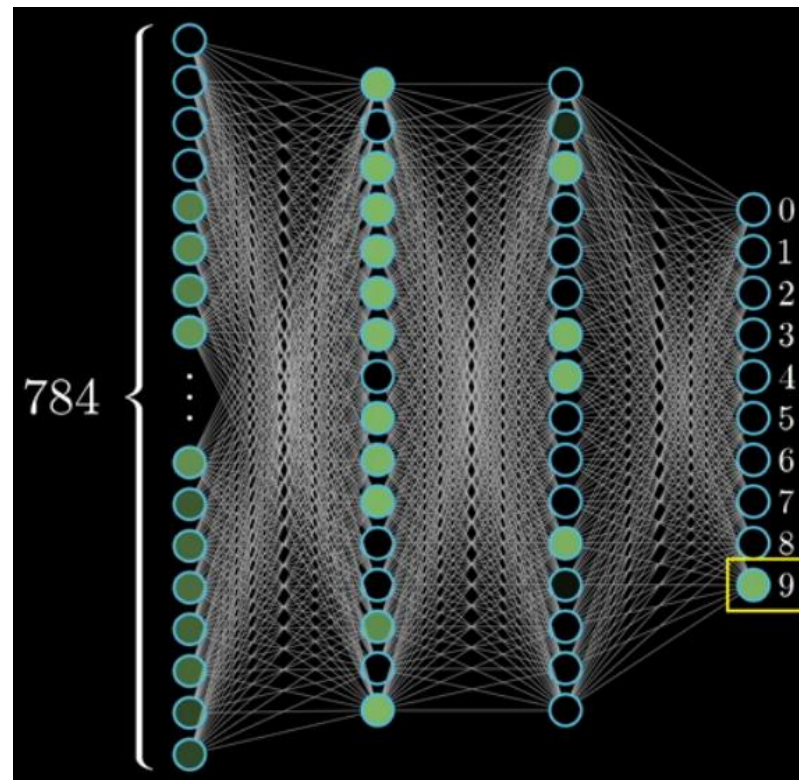
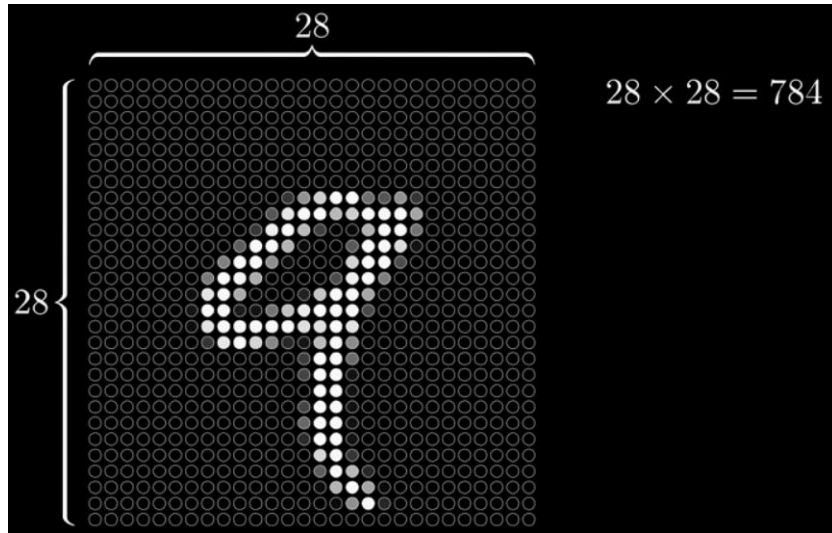
1. Analytical method
2. Learning Rule

Analytical method – It is tenable only for small number of inputs, for large number of inputs the method is not practical.

The use of **learning rule** or **algorithm** is pertinent with perceptron having many inputs or neural network with many layers.

ANN

<https://www.youtube.com/watch?v=aircAruvnKk&t=990s>



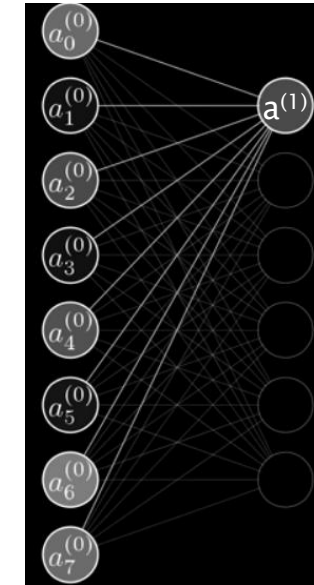
$$784 \times 16 + 16 \times 16 + 16 \times 10$$

weights

$$16 + 16 + 10$$

biases

$$13,002$$



1st neuron
of 2nd Layer

$$\mathbf{a}^{(1)} = \sigma(\mathbf{W}\mathbf{a}^{(0)} + \mathbf{b})$$

$$\sigma \left(\begin{bmatrix} w_{0,0} & w_{0,1} & \dots & w_{0,n} \\ w_{1,0} & w_{1,1} & \dots & w_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{k,0} & w_{k,1} & \dots & w_{k,n} \end{bmatrix} \begin{bmatrix} a_0^{(0)} \\ a_1^{(0)} \\ \vdots \\ a_n^{(0)} \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{bmatrix} \right)$$

*A Degree is not a Degree in Action,
If not for True Learning and Function.*

Thank You